In such nuclear reactions it frequently happens that unstable atoms are produced; radioactive atoms, not occurring in nature, which change after a certain length of time into some stable atomic species. In the cases hitherto known, this process occurs with emission of electrons or positrons only. This fact completes our survey of the nature of the particles which issue forth from atomic nuclei, whether by spontaneous emission or due to outside interference.

III. THE BUILDING BLOCKS OF ATOMIC NUCLEI

With the knowledge which we have gained we can now proceed to consider the question, which of the elementary particles can be regarded as the ultimate building blocks of atomic nuclei. Let us, first of all, enumerate these particles once again? The *proton* and the *neutron*, the *electron* and the *positron*, the *neutrino*, and the *gamma-ray photon.* There are, furthermore, the *alpha particles.* But in view of the mass and charge numbers of the latter, we may assume that they are not elementary particles at all, but composite structures. However, our list of elementary particles is still incomplete. In the first place, it does not include the *antineutrino*, which is the opposite number of the neutrino, as the positron is the opposite number of the electron. While it has the same near-zero mass and zero charge, it differs from the neutrino as regards a property which is present in all elementary particles, and which we have not mentioned as yet: Its *spin* (or as it is called technically, its *angular momentum*) is opposite in direction for a given direction of the magnetic moment. Many elementary particles behave mechanically like small spinning tops. But their angular momenta can have only certain definite values, which can be accounted for by quantum mechanics. In the cases of the elementary particles with which we are concerned here, this value is, generally, $\hbar/2$ or h; this $\hbar$ is an abbreviated symbol for $h/2\pi$, and h is Planck's constant. As a general rule, the effect of this spin moment is that the particles possess a magnetic moment, in other words, they act like small magnets. In the case of heavier particles, this magnetic moment is measured in terms of a unit called *nuclear magneton* (*n.m.*), while for lighter particles a larger unit, called *Bohr magneton* (*B.m.*) is used. (See Table I.)

There is still another elementary particle we must mention; it is called *meson*, for its mass is between that of an electron and that of a proton (the Greek word 'meson' means 'the middle one'). Mesons can be observed in cosmic radiation, and before we go into further details about their properties, we must first say a few words about the nature of cosmic radiation.

As a result of the studies of Hess and Kohlhörster, we have known for almost forty years that a faint, continuous, extremely penetrating radiation